

Lucas Narloch Zabla

Backend Developer | Python & FastAPI

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Backend Developer with production experience developing Python and FastAPI applications, RESTful APIs, internal tools, and automation solutions in AWS environments. Experienced in integrating backend services with relational databases, corporate systems, IoT devices, and external platforms. Focused on clean and maintainable code, automated testing, logging, troubleshooting, infrastructure automation, and continuous improvement.

TECHNICAL SKILLS

- **Backend:** Python, FastAPI, Node.js, TypeScript, NestJS, RESTful APIs, Object-Oriented Programming
- **Databases:** PostgreSQL, MySQL, DynamoDB
- **Testing & Quality:** pytest, Testcontainers, unit testing, integration testing, error handling, logging
- **Cloud & DevOps:** AWS Lambda, S3, SQS, EC2, RDS, IoT Core, CloudWatch, Terraform, GitHub Actions, Docker
- **Security:** JWT, OAuth 2.0, OpenID Connect, AWS Cognito
- **Tools:** Git, Linux, Bash, Nginx, Prometheus, Grafana

EXPERIENCE

Backend Developer - SmartLy Fabricação de Dispositivos Inteligentes Ltda. 10/2026 - Present

- Lead the consolidation of multiple AWS Lambda functions into a single FastAPI API organized into services and repositories, reducing deployment time from approximately 12 to 5 minutes and latency on several endpoints from 7–10 seconds to approximately 1.5 seconds (79–85%).
- Redesigned AWS network architecture, replacing multiple VPC endpoints with a NAT Gateway and adding an EC2 bastion for database access, reducing monthly network costs by approximately 64% (US\$90 to US\$32) and expanding Lambda connectivity.
- I am restructuring the MySQL schema of a production IoT platform, remodeling entities, relationships, and migration flows to better reflect the domain and preserve integrity and compatibility during transition.

Backend Developer Intern - SmartLy Fabricação de Dispositivos Inteligentes Ltda. 10/2024 - 09/2026

- Developed an endpoint for data migration between IoT devices, integrating MySQL, DynamoDB, S3, and AWS IoT Core via MQTT, supporting thousands of historical records per operation.
- Implemented an automated test suite for FastAPI APIs with pytest and Testcontainers, including 120 integration tests and 300 unit tests, improving delivery reliability and reducing production regressions.
- Analyzed Lambda error metrics in CloudWatch and fixed unhandled edge cases, reducing average errors from 113 to 5 per five-minute interval.

PROJECTS

Sunlog.dev

Production devlog platform with REST APIs, PostgreSQL, JWT, asynchronous processing, and automated AWS infrastructure. Includes 33 HTTP endpoints, 100+ tests, GitHub Actions CI/CD, Docker, Nginx, and Prometheus/Grafana observability.

Technologies: NestJS, TypeScript, PostgreSQL, JWT, Docker, Nginx, AWS, GitHub Actions, Prometheus, Grafana, Next.js

[Code](#) | [Live demo](#)

PocScript

Programming language and compiler implemented in C, covering the full compilation pipeline: lexical analysis, AST parsing, semantic analysis, custom IR generation, LLVM IR emission, and native executable generation. Includes a custom runtime based on Linux syscalls, static type system, scope resolution, arrays, pointers, error handling, and a suite with 196 unit and integration tests.

Technologies: C, LLVM IR, NASM, Make, Linux syscalls, Compiler Design

[Code](#)

EDUCATION

Bachelor of Computer Science - PUCPR 02/2025 - 12/2028

Full scholarship (100%). Relevant coursework: algorithms, data structures, OOP, operating systems, networks, databases, and software engineering.

Technical High School in Electronics - CEEP 02/2022 - 12/2024

LANGUAGES

- **Portuguese:** Native
- **English:** Professional Working Proficiency